Report of the Paragraph Classification of Jin Yong's Novels Based on LDA Topic Modeling

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Abstract

This is a Report of the Paragraph Classification of Jin Yong's Novels Based on LDA Topic Modeling. It employs the Latent Dirichlet Allocation (LDA) model in conjunction with a Support Vector Machine (SVM) classifier to conduct thematic modeling and classification experiments on novel paragraphs. By adjusting parameters such as the number of topics and paragraph length, the study systematically investigates the impact of various variables on classification performance. The experimental data indicate that : there exists an optimal range for the number of topics; the performance of classification under the word-based segmentation significantly outperforms character-based segmentation under the same number of topics and paragraph length. What's more, regardless of whether words or characters are used as the basic units, there is a significant positive correlation between paragraph length and mean classification accuracy. The highest accuracy rate of 98.1% is achieved with a configuration based on basic units of words and 10 topics when the paragraph length reaches 3000 words.

Introduction

Text classification and topic modeling are fundamental tasks in natural language processing (NLP), with applications ranging from document organization to semantic analysis. In this study, we focus on the works of Jin Yong, a renowned Chinese novelist, to explore the effectiveness of Latent Dirichlet Allocation (LDA) and Support Vector Machine (SVM) in modeling and classifying Chinese literary texts. By uniformly sampling 1,000 paragraphs from his novels and varying key parameters — such as the number of topics (T), the basic units ("word" vs. "character"), and paragraph length (K) — we aim to systematically evaluate the performance of LDA in capturing semantic structures and the impact on classification accuracy. Using 10-fold cross-validation, we assess the model's robustness and provide insights into optimal configurations for practical text analysis tasks. This work not only enhances the understanding of topic modeling in Chinese literature but also offers data-driven recommendations for model selection and optimization in similar contexts.

Methodology

1. Latent Dirichlet Allocation

Latent Dirichlet Allocation (LDA) constitutes a probabilistic generative model for text corpora that operates under the fundamental assumption that documents manifest as mixtures of latent topics, where each topic is characterized by a probability distribution over words. This hierarchical Bayesian framework employs Dirichlet priors to model document-topic (θ_d) and topic-word (φ_k) distributions, thereby establishing a complete data generation mechanism.

The generative process of LDA unfolds through the following probabilistic steps: For each document d in the corpus, the model first draws a topic proportion vector θ_d from a Dirichlet distribution with parameter α, expressed as $\theta_d \sim Dir(\alpha)$. Concurrently, for each topic k, it generates a word distribution φ_k from another Dirichlet distribution with parameter η, denoted as $\varphi_k \sim Dir(\eta)$. When generating the nth word in document d, the model initially samples a topic assignment $z_{d,n}$ from the document-specific topic distribution $\left(z_{d,n} \sim Multinomial(\theta_d)\right)$,

subsequently drawing the observed word $w_{d,n}$ from the corresponding topic's word distribution $\left(w_{d,n} \sim Multinomial(\varphi_{z_{d,n}})\right)$.
The complete joint probability distribution of the model factors as:
$P(W, Z, \theta, \phi|\alpha, \eta) = \prod_{d=1}^{D} P(\theta_d|\alpha) \prod_{k=1}^{K} P(\phi_k|\eta) \prod_{n=1}^{N_d} P(z_{d,n}|\theta_d) P(w_{d,n}|\phi_{z_{d,n}})$
This factorization explicitly represents the hierarchical dependencies between parameters and observed variables. The Dirichlet hyperparameters α and η exert crucial control over the sparsity patterns in topic distributions, where smaller values of α encourage documents to concentrate on fewer topics, while reduced η values promote topics dominated by fewer characteristic words.
The generative process finds its formal representation through the plate notation: $\alpha \to \theta_d \to z_{d,n} \to w_{d,n} \leftarrow \varphi_k \leftarrow \eta$, where α and η denote hyperparameters, θ_d and φ_k represent latent variables, $z_{d,n}$ corresponds to topic assignments, and $w_{d,n}$ signifies observed word tokens. This graphical model elegantly captures the conditional independence relationships inherent in the LDA framework.
As a fully generative model, LDA provides not only a principled approach to document modeling but also establishes the theoretical foundation for subsequent variational inference or Gibbs sampling procedures. The model's capacity to uncover latent semantic structures while maintaining computational tractability has rendered it particularly valuable for diverse applications including topic discovery, document classification, and information retrieval within the computational linguistics and text mining communities.

2. Support Vector Machine
The Support Vector Machine (SVM) is a supervised learning algorithm designed for classification and regression tasks. Its core principle is to identify an optimal hyperplane in a high-dimensional feature space that maximally separates data points of different classes. For linearly separable data, this hyperplane is derived by solving a convex optimization problem that maximizes the margin—the distance between the hyperplane and the nearest data points (support vectors) from each class.

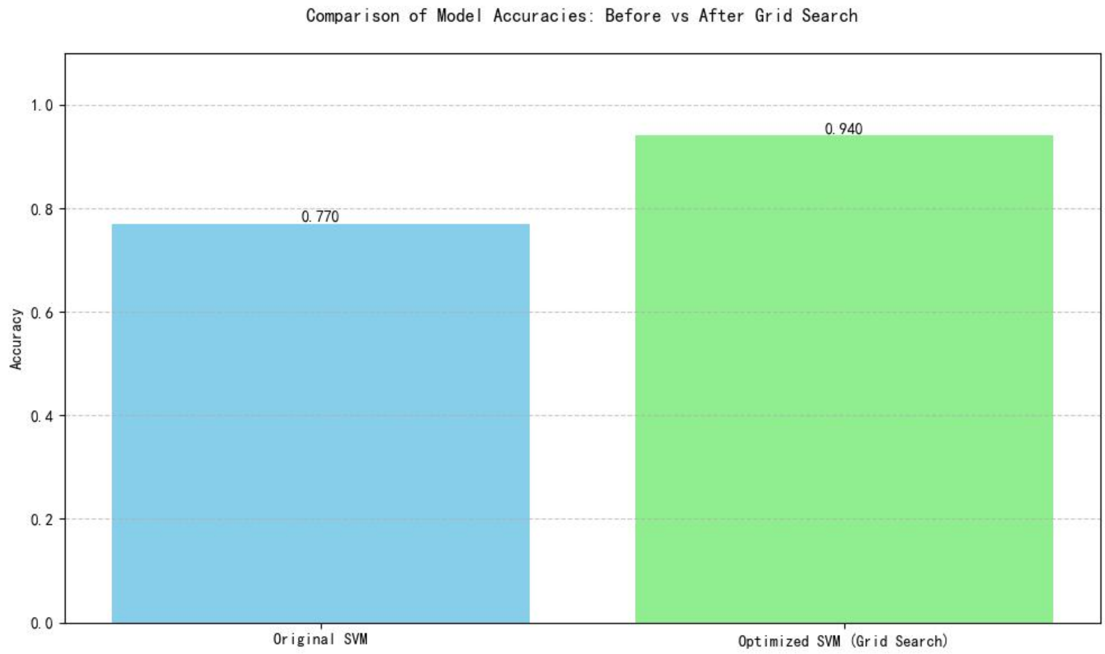
The experiment in this study developed a modular text classification system combining a Latent Dirichlet Allocation (LDA) topic model and a Support Vector Machine (SVM) classifier. The implementation began by initializing global parameters, including paragraph length and a predefined list of topic numbers (T), alongside integrating memory monitoring tools to track computational resource utilization. A core cross-validation function was designed to encapsulate the entire workflow: LDA model training for topic distribution extraction, transformation of text segments into topic probability vectors, SVM classifier construction, and performance evaluation. Additionally, a dedicated function was implemented to systematically process each topic number configuration. The main execution strategy adopted a progressive approach: preliminary small-scale tests with reduced parameter sets (e.g., shorter paragraphs, fewer topics) were conducted to validate system stability and workflow feasibility, followed by parallelized large-scale experiments to evaluate performance across all predefined T values. For each topic number setting, a rigorous 10-fold cross-validation procedure was applied, generating the mean classification accuracy and standard deviation as key metrics. The system architecture emphasized modularity, robustness, and reproducibility, incorporating real-time resource monitoring to prevent memory overflow and consistency checks to ensure algorithmic reliability. This design not only enhanced debuggability during development but also facilitated scalable extensions for future text analysis tasks.
The experiment comprises three key steps: data preprocessing, model construction and training, and model evaluation.

Step 1: Data Preprocessing:

From a designated corpus, 1,000 paragraphs were uniformly sampled, and the text data from the txt files were read and stored as lists. For character-level experiments, punctuation and stopwords were first removed from the text data, after which each Chinese string in the list was segmented into individual Chinese characters and stored as a new list. For word-level experiments, punctuation was first eliminated, followed by Chinese word segmentation using the jieba library; subsequently, stopwords were removed, and the processed text was stored as a new list.

Step2: Model Construction and Training

The Latent Dirichlet Allocation (LDA) model was employed to generate topic distributions. The segmented text fragments were transformed into LDA topic distribution vectors, and new text fragments were converted into LDA features for prediction using a Support Vector Machine (SVM) classifier as the classification model. When the paragraph length (K) was fixed at 1,000, the number of topics (T) was set to 5, 10, 20, 40, and 80, respectively. Conversely, when the number of topics was fixed at 20, the paragraph lengths were set to 20, 100, 500, 1,000, and 3,000. During model training, grid search was utilized to determine optimal parameters, which were then used as the initial parameters for the SVM model to enhance accuracy. Additionally, a progressive testing strategy was implemented—small K values (20, 100) were tested first to verify the workflow's feasibility, and the full parameter set was executed only after successful validation of these smaller values, thereby improving code debuggability.



Step 3: Model Evaluation:

A 10-fold cross-validation was conducted, and the average accuracy along with its standard deviation were computed to assess the classification performance of the model.

Experimental Studies

This study employs a controlled variable approach to systematically investigate the impact of different parameters on classification performance, with the experimental design encompassing three dimensions:

1. **Topic Quantity Dimension:**
With paragraph length fixed at K = 1000, the influence of topic numbers ($T \in \{5, 10, 20, 40, 80\}$) on model accuracy and stability is systematically evaluated under both word-based and character-based segmentation.
2. **Paragraph Length Dimension:**
With the topic number fixed at T = 10, classification performance is measured across varying paragraph lengths ($K \in \{20, 100, 500, 1000, 3000\}$) using 10-fold cross-validation (reported as mean accuracy ± standard deviation) for both basic units of words and characters.
3. **Text Granularity Dimension:**
Under identical configurations of paragraph length and topic number, the classification performance of basic units of words is compared with that of basic units of characters.

Performance Under Word-Based Segmentation (Fixed $T = 10$, Varying K)

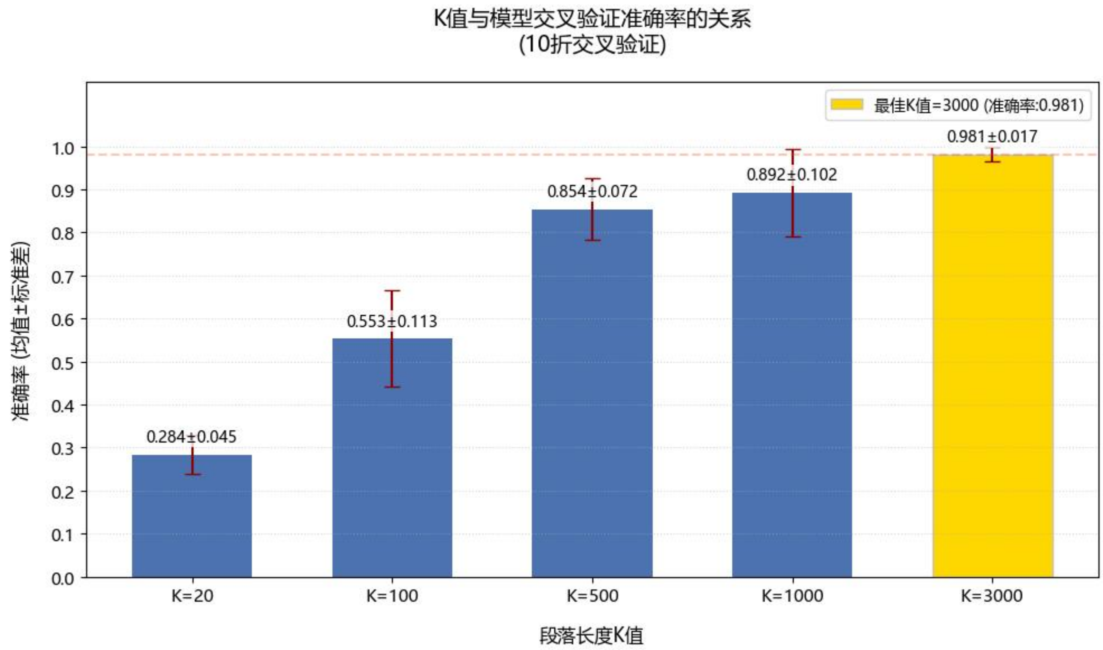


Figure 1： The mean accuracy and standard deviation across varying values of K (word)

As illustrated in Figure 1, when the number of topics is 10, the classification performance significantly improves with longer paragraphs, such as K=3000, achieving an average accuracy rate of 0.981±0.017 in 10-fold cross-validation, whereas the accuracy rate for short paragraphs, such as K=20, is only 0.284±0.045. This discrepancy may stem from the richer context information provided by longer paragraphs, which allows LDA to capture a more stable and coherent topic distribution, thereby providing the SVM classifier with more discriminative feature vectors. Additionally, the standard deviation significantly decreases with longer paragraphs (e.g., 0.017 for K=3000 and 0.102 for K=100), indicating that longer texts reduce the volatility in topic modeling due to data sparsity, thus enhancing the robustness of the model.

Performance Under Word-Based Segmentation (Fixed K = 1000, Varying T)

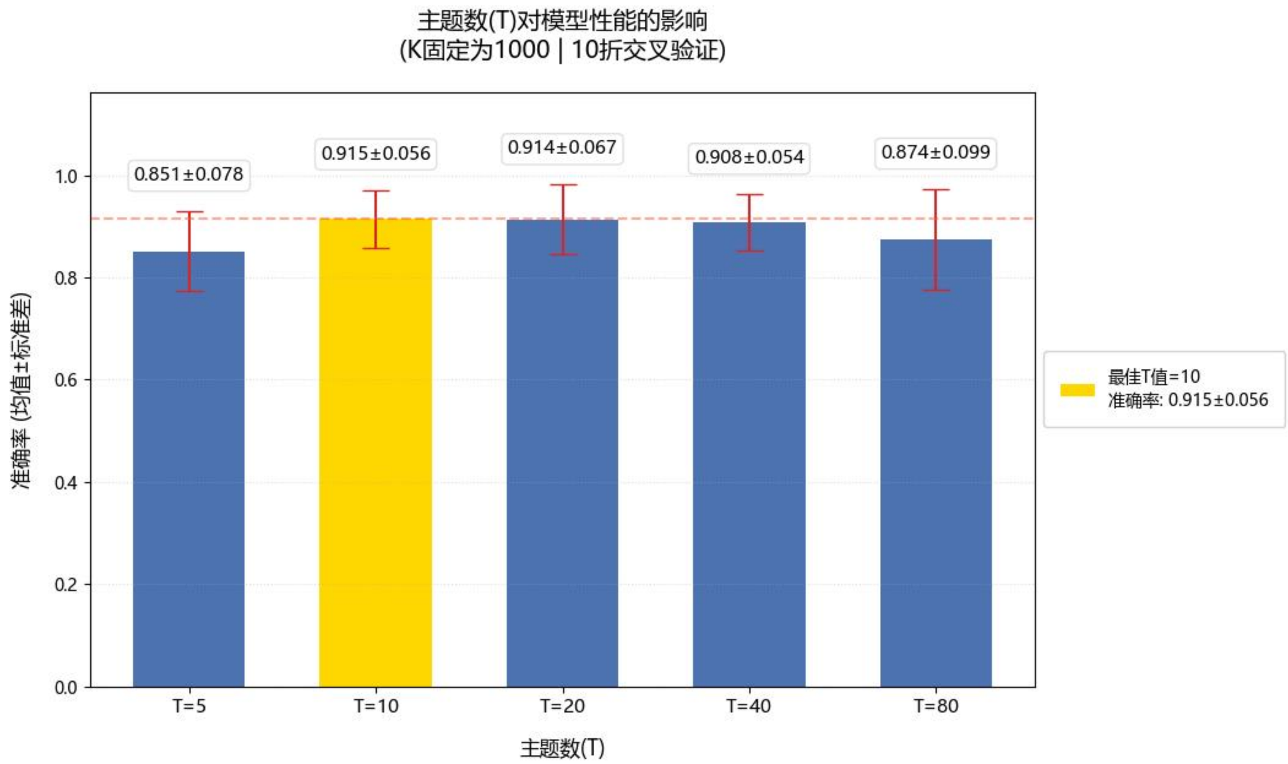


Figure 2： The mean accuracy and standard deviation across varying values of T (word)

The results in Figure 2 demonstrate that the model’s accuracy rate reaches its peak at 0.915±0.056 when T=10, indicating that T=10 is the optimal choice of topic number in this specific experimental setup. The chart also displays the error range (standard deviation) for each topic number. For instance, the error range for T=10 is ±0.056, suggesting that the variation in accuracy rate across multiple experiments is relatively small, and the model's performance is relatively stable. Overall, the model's accuracy rate first increases and then decreases with the increase in the number of topics. An insufficient number of topics may prevent the model from adequately capturing the complexity of the data, while an excessive number of topics may lead to overfitting, thus reducing the model's generalization ability on unseen data.

Performance Under Character-Based Segmentation (Fixed T = 10, Varying K)

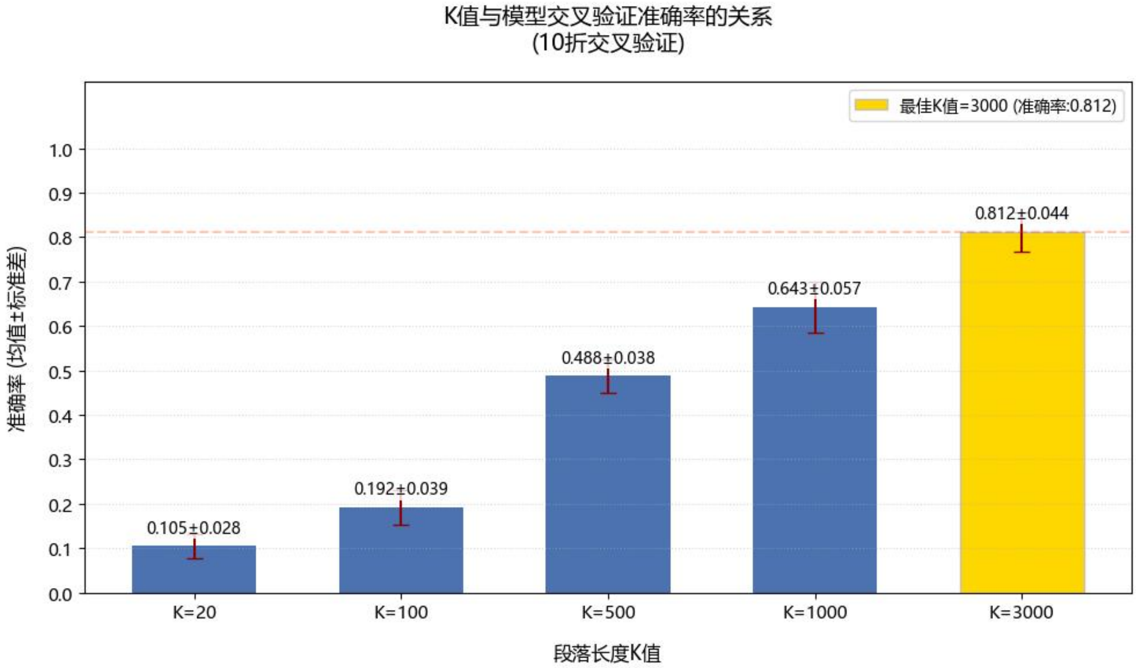


Figure 3： The mean accuracy and standard deviation across varying values of K (character)

As shown in Figure 3, for basic units of character, the model's accuracy rate gradually improves with the increase in paragraph length (K). Shorter paragraph lengths may prevent the model from fully capturing the complexity of the data, while longer paragraphs may provide more context information, allowing the model to better learn and distinguish between subtle differences between different categories. By comparing the experimental results of "basic units of words" and "basic units of characters," it can be observed that under the same number of topics and paragraph lengths, the classification performance of basic units of words is significantly higher than that of baisc units of characters. This gap may arise from the fact that basic units of words preserve semantic integrity, enabling LDA to capture coherent topic distributions, while basic units of characters break down the text into independent characters, weakening the SVM's discriminative power.

Performance Under Character-Based Segmentation (Fixed K = 1000, Varying T)

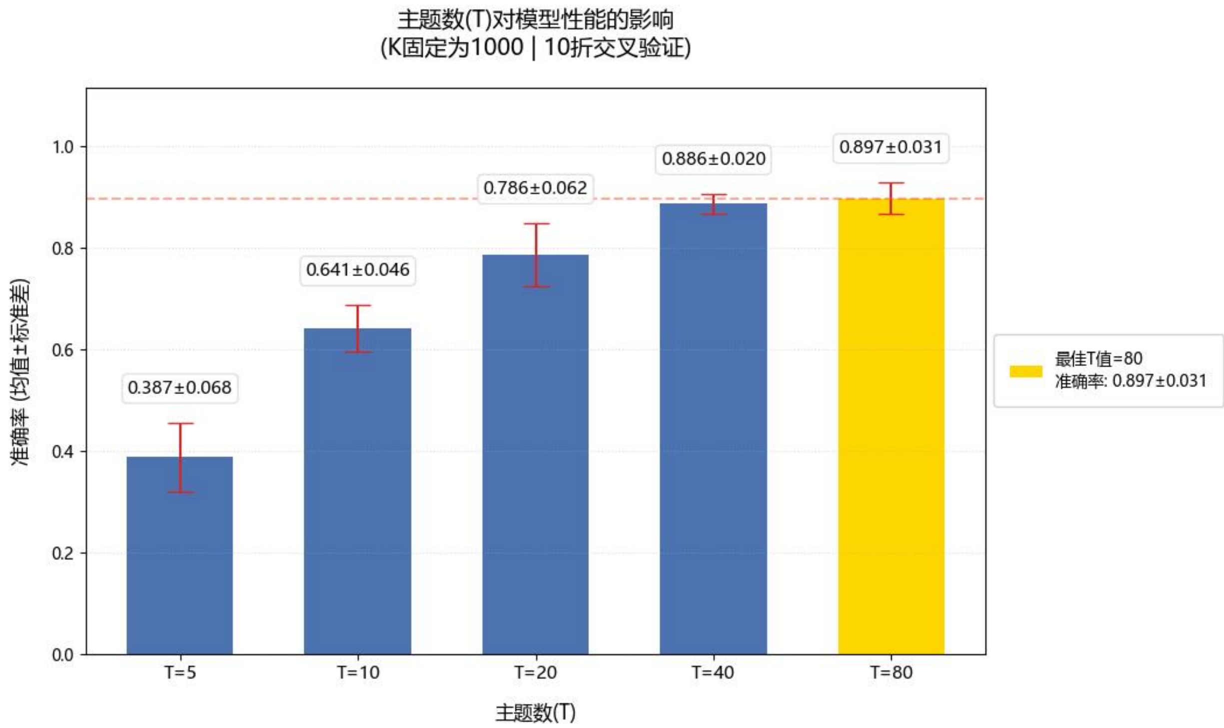


Figure 4： The mean accuracy and standard deviation across varying values of T (character)

The figure shows that with the increase in the number of topics (T), the model's accuracy rate generally trends upward. This indicates that a greater number of topics can help the model better capture the complexity and diversity of the data. By comparing the experimental results of "basic units of words" and "basic units of characters," we can similarly observe that under the same conditions, the classification performance of word units is significantly higher than that of basic units of characters.

Conclusions

This study systematically investigates the impact of key parameters—topic quantity (T), paragraph length (K), and text granularity units (word vs. character)—on the performance of an LDA-SVM text classification framework. Through controlled experiments and quantitative analysis, the following conclusions are drawn:

Firstly, for word-based segmentation, moderate topic numbers (T = 10) achieve peak accuracy (e.g., 0.915 ± 0.056 at T = 10, K = 1000), balancing semantic granularity and noise suppression. Excessive topics (T ≥ 40) fragment coherent themes, degrading generalization (e.g., 0.874 ± 0.099 at T = 80), while insufficient topics (T ≤ 5) oversimplify semantic structures (0.051 ± 0.078 at T = 5). At the same time, longer texts (K ≥ 1000) significantly enhance classification performance (e.g., 0.981 ± 0.017 at K = 3000 vs. 0.284 ± 0.045 at K = 20), as they provide sufficient contextual information for LDA to model stable topic distributions. The reduced standard deviation (e.g., 0.017 at K = 3000) further validates the robustness of long-text modeling.

Secondly, a significant positive correlation exists between paragraph length and classification accuracy across both word-based and character-based segmentation units. For instance, when K reaches 3,000 words, the configuration combining basic units of words and $T = 10$ achieves the highest accuracy of 98.1%.

Thirdly, word-based segmentation consistently outperforms basic units of characters across all configurations. For instance, at K = 1000 and T = 10, basic units of words achieve 0.915 ± 0.056, whereas basic units of characters attain only 0.641 ± 0.046. This disparity arises from the semantic integrity preserved by basic units of words, which enables LDA to generate interpretable topic distributions, while character units fragment text into noise-prone symbols. What's more, for applications involving long texts ($K \geq 1000$), word-based segmentation paired with moderate topic numbers ($T = 10$–20) is strongly recommended to maximize accuracy and stability.

In summary, this work provides empirical evidence that semantic coherence and parameter synergy are pivotal for optimizing LDA-SVM frameworks. The conclusions offer actionable insights for text classification tasks, bridging theoretical analysis with practical implementation.